

Luna Dales

Nottingham, NH

Class of 2027 | Virtual Learning Academy Charter School, Exeter, NH

EDUCATION

Virtual Learning Academy Charter School, Exeter, NH

High School Diploma, Expected 2027

Cumulative GPA: 4.00 / 4.00 | Designation: NH Scholar

SAT: 1520 (770 Math, 750 Reading and Writing)

Intended Major: Aerospace Engineering | Additional Interest: Computer Science

Advanced Coursework

- Completed AP courses: Physics 1; Computer Science Principles
- In progress: AP Computer Science A; Precalculus Honors
- Honors mathematics and science: Algebra 1 and 2; Geometry; Biology; Chemistry; Physical Science
- Additional coursework: English 1-3 Honors; U.S. History Honors; Mandarin Chinese 1-3, including Mandarin Chinese 3 Honors; Creative Coding JavaScript 1

ENGINEERING AND TECHNOLOGY PROJECTS

Accessible Shopping App | Independent Project | Summer 2026-Present

- Planning an iPhone app to help people with low vision locate products and navigate unfamiliar stores; developed preliminary app designs and layouts.
- Researched product-location information, store maps, public API availability, and planograms across Publix, Hannaford, Shaw's, Lowe's, Home Depot, and Walmart.
- Revised the design to accommodate differences in the location information available from each store chain.
- Sought input from NOAH peers and discussed opportunities to help or test the app; incorporated in-store navigation into the design following feedback.
- Exploring camera assistance for identifying products on shelves and an AI search assistant; these features remain in the planning stage.

Rocket Project | Independent Project

- Develop a flight computer through component selection, schematic and printed circuit board (PCB) design, assembly, testing, and software development.
- Develop and test thrust vector control (TVC) software and guidance functionality.
- Design and build rockets; use MATLAB, Simulink, and OpenRocket for modeling, simulation, and analysis.
- Apply soldering, PCB assembly, and 3D printing skills to project hardware.

VOLUNTEER EXPERIENCE

Crocheter and Volunteer | Warm Up America Foundation

- Crocheted 50 blanket squares, 10 hats, and 10 pairs of gloves for donation to people in need, as well as many amigurumi animals.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Martial Arts | Student and Peer Instructor | High School

- Train in two martial arts disciplines at different dojos, progressing through several belts and practicing with both children and adults.
- Help teach younger students by explaining and demonstrating techniques and recognizing when a different approach is needed.
- Develop situational awareness, reflexes, and adaptive techniques to compensate for limited depth perception.

National Organization for Albinism and Hypopigmentation | NOAH

- Led an ongoing teen book club.
- Crocheted a giveaway prize for a fundraiser.
- Participate in annual national conferences.
- Co-led a low-vision sailing workshop.

ADDITIONAL EXPERIENCES AND PROGRAMS

Space Camp for Interested Visually Impaired Students | SCIVIS

- Served as Commander, Mission Specialist, and Flight Director in simulated International Space Station missions.
- Participated in adapted astronaut training, including the Multi-Axis Trainer, 5 Degrees of Freedom Chair, and Underwater Astronaut Trainer.
- Developed an interest in designing spacecraft and interfaces that accommodate a broader range of human abilities.

Summer Academic and Leadership Programs

- Coding Academy: Intro to Python | Held at Harvard University
- Business Academy: Entrepreneurship | Held at Harvard University
- National Student Leadership Conference: Aerospace Program | Held at Duke University

Family Responsibilities | Freshman and Sophomore Years

- Shared household and family caregiving responsibilities with an older sister during extended parental illnesses while maintaining academic performance.

AWARDS AND CERTIFICATIONS

- Level 1 High Power Rocketry Certification | National Association of Rocketry (NAR)
- College Board National Recognition Program - Rural and Small-Town Recognition Award (2026)
- NH Scholar designation, as recorded on the VLACS transcript

SKILLS AND INTERESTS

- Hardware: soldering; schematic and PCB design, assembly, and testing; 3D printing
- Engineering software: MATLAB, Simulink, and OpenRocket for simulation and analysis
- Programming coursework: AP Computer Science Principles; Creative Coding JavaScript 1; AP Computer Science A in progress
- Accessibility research, preliminary app layout design, and user feedback gathering
- Peer instruction, collaboration, and adaptive problem-solving
- Aerospace engineering, human factors, accessible spaceflight, and martial arts